SQL FOR DATA ANALYSIS · LESSON 6 OF 15

SQL Functions: Aggregate, Text, Date & Number

Use SQL Server's built-in functions to summarise, clean and reshape data. Aggregate functions (COUNT, SUM, AVG, MIN, MAX, STRING_AGG), text functions (LEN, LEFT, RIGHT, SUBSTRING, CHARINDEX, REPLACE, TRIM, UPPER, CONCAT), date functions (GETDATE, YEAR, MONTH, DATEPART, DATENAME, DATEADD, DATEDIFF, EOMONTH) and number functions (ROUND, CEILING, FLOOR, ABS, POWER, SQRT), plus CAST and CONVERT.

SQL class 4 · SQL Server 2022, 2026 · Beginner–Intermediate · 120 min

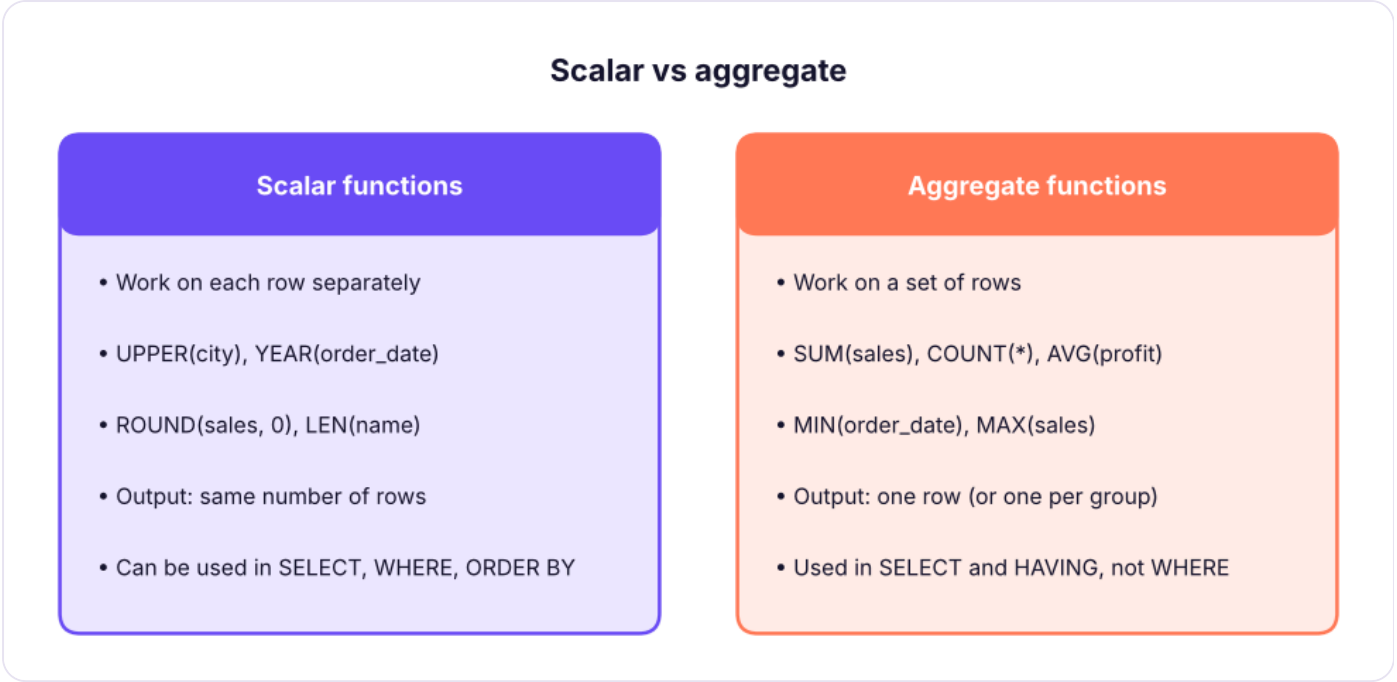
By the end of this lesson you can

- Distinguish scalar functions (one value per row) from aggregate functions (one value per group)
- Summarise data with COUNT(*), COUNT(column), COUNT(DISTINCT), SUM, AVG, MIN, MAX and STRING_AGG
- Clean and extract text with LEN, LEFT, RIGHT, SUBSTRING, CHARINDEX, REPLACE, TRIM, UPPER, LOWER and CONCAT
- Work with dates using YEAR, MONTH, DATEPART, DATENAME, DATEADD, DATEDIFF, EOMONTH and DATEFROMPARTS
- Round and convert numbers with ROUND, CEILING, FLOOR, ABS, POWER, SQRT, CAST, CONVERT and TRY_CAST

Datasets used

`superstore_sales.csv`

Two kinds of functions



Scalar functions return one value per row. Aggregate functions collapse many rows into one value (per group).

Aggregate functions

Function	Returns	NULL handling
`COUNT(*)`	Number of rows	Counts every row
`COUNT(col)`	Number of non-NULL values	Skips NULLs
`COUNT(DISTINCT col)`	Number of distinct non-NULL values	Skips NULLs
`SUM(col)`	Total	Skips NULLs
`AVG(col)`	Average of non-NULL values	Skips NULLs (they are not treated as 0)
`MIN(col)` , `MAX(col)`	Smallest / largest (numbers, text, dates)	Skips NULLs
`STDEV(col)` , `VAR(col)`	Sample standard deviation / variance	Skips NULLs
`STRING_AGG(col, sep)`	Values joined into one string	Skips NULLs

SQL

```sql
SELECT COUNT(*)                       AS order_lines,
       COUNT(DISTINCT order_id)       AS orders,
       COUNT(DISTINCT customer_id)    AS customers,
       ROUND(SUM(sales), 0)           AS total_sales,
       ROUND(AVG(sales), 0)           AS avg_line_sales,
       MIN(order_date)                AS first_order,
       MAX(order_date)                AS last_order,
       ROUND(STDEV(sales), 0)         AS sd_sales
FROM superstore;
```

RESULT

order_lines	orders	customers	total_sales	avg_line_sales	first_order	last_order
6323	3200	785	201,413,569	31,854	2023-01-01	2025-12-31

1 rows × 8 columns

SQL

```sql
SELECT category,
       STRING_AGG(sub_category, ', ') AS sub_categories
FROM (SELECT DISTINCT category, sub_category FROM superstore) AS t
GROUP BY category
ORDER BY category;
```

RESULT

category	sub_categories
Furniture	Tables, Furnishings, Bookcases, Chairs
Office Supplies	Storage, Appliances, Art, Binders, Paper
Technology	Copiers, Phones, Machines, Accessories

3 rows × 2 columns

In SQL Server add `WITHIN GROUP (ORDER BY sub_category)` after `STRING_AGG(...)` to control the order of the joined values.

WATCH OUT **AVG of integers**

`AVG` on an `INT` column returns an integer in SQL Server (`AVG` of 1, 2 = 1). Use `AVG(quantity * 1.0)` or `AVG(CAST(quantity AS DECIMAL(10,2)))` for decimals.

Text (string) functions

Function	Example	Result
`LEN(s)`	`LEN('Pune ')`	4 (trailing spaces not counted)
`UPPER(s)`, `LOWER(s)`	`UPPER('pune')`	PUNE
`LEFT(s, n)`, `RIGHT(s, n)`	`LEFT('IN-2025-00042', 2)`	IN
`SUBSTRING(s, start, n)`	`SUBSTRING('IN-2025-00042', 4, 4)`	2025
`CHARINDEX(find, s)`	`CHARINDEX('-', 'IN-2025-00042')`	3
`REPLACE(s, old, new)`	`REPLACE('2025-04-27', '-', '/')`	2025/04/27
`TRIM(s)`, `LTRIM`, `RTRIM`	`TRIM(' Delhi ')`	Delhi
`CONCAT(a, b, …)`	`CONCAT(city, ', ', state)`	Pune, Maharashtra
`REVERSE(s)`	`REVERSE('abc')`	cba
`REPLICATE(s, n)`	`REPLICATE('*', 3)`	***
`STRING_SPLIT(s, sep)`	`SELECT value FROM STRING_SPLIT('a,b,c', ',')`	3 rows
`FORMAT(value, fmt)`	`FORMAT(1234567, 'N0', 'en-IN')`	12,34,567

SQL

```sql
SELECT TOP 6
       order_id,
       LEFT(order_id, 2)                                AS country,
       SUBSTRING(order_id, 4, 4)                        AS order_year,
       RIGHT(order_id, 5)                               AS order_no,
       product_name,
       LEFT(product_name, CHARINDEX(' ', product_name) - 1) AS brand,
       UPPER(city)                                      AS city_upper,
       LEN(customer_name)                               AS name_length
FROM superstore
ORDER BY row_id;
```

RESULT

order_id	country	order_year	order_no	product_name	brand	city_upper	name_
IN-2023-00001	IN	2023	00001	Nova Phone 223	Nova	BENGALURU	12
IN-2023-00001	IN	2023	00001	Orbit Appliance 798	Orbit	BENGALURU	12
IN-2023-00001	IN	2023	00001	Orbit Table 248	Orbit	BENGALURU	12
IN-2023-00002	IN	2023	00002	Crest Storage 946	Crest	AHMEDABAD	11
IN-2023-00003	IN	2023	00003	Crest Bookcase 170	Crest	AHMEDABAD	10
IN-2023-00003	IN	2023	00003	Zenith Table 111	Zenith	AHMEDABAD	10

6 rows × 8 columns

The brand is everything before the first space: `CHARINDEX(' ', product_name)` finds the space position, and `LEFT(…, position - 1)` cuts there. This is the SQL version of Excel's `LEFT(A2, FIND(" ", A2) - 1)` from Excel Lesson 5.

Cleaning text

```sql
WITH raw AS (
    SELECT '  bengaluru ' AS city UNION ALL
    SELECT 'BENGALURU'          UNION ALL
    SELECT 'Bangalore'          UNION ALL
    SELECT 'Bengaluru'
)
SELECT city                                                    AS raw_value,
       CONCAT(UPPER(LEFT(TRIM(city), 1)), LOWER(SUBSTRING(TRIM(city), 2, 100))) AS
proper_case,
       CASE WHEN LOWER(TRIM(city)) IN ('bangalore', 'bengaluru') THEN 'Bengaluru' ELSE
TRIM(city) END AS standardised
FROM raw;
```

RESULT

raw_value	proper_case	standardised
bengaluru	Bengaluru	Bengaluru
BENGALURU	Bengaluru	Bengaluru
Bangalore	Bangalore	Bengaluru
Bengaluru	Bengaluru	Bengaluru

4 rows × 3 columns

(SQL Server has no `PROPER` function; joining `UPPER(LEFT(…,1))` and `LOWER(SUBSTRING(…,2,…))` capitalises the first letter of a single word. Standardising spellings is best done with `CASE` or a mapping table joined in.)

> **NOTE** **`+` vs CONCAT for text**
>
> In SQL Server `'a' + 'b'` joins text, but `'a' + NULL` is NULL and `'Q' + 2` fails (it tries to convert 'Q' to a number). `CONCAT('Q', 2)` gives Q2 and treats NULL as empty. Prefer `CONCAT`.

Date and time functions

Function	Example (d = '2026-04-27')	Result
`GETDATE()`, `SYSDATETIME()`		Current date and time
`CAST(GETDATE() AS DATE)`		Today's date
`YEAR(d)`, `MONTH(d)`, `DAY(d)`	`MONTH(d)`	4
`DATEPART(part, d)`	`DATEPART(quarter, d)`	2
`DATENAME(part, d)`	`DATENAME(month, d)`, `DATENAME(weekday, d)`	April, Monday
`DATEADD(part, n, d)`	`DATEADD(month, 3, d)`	2026-07-27
`DATEDIFF(part, start, end)`	`DATEDIFF(day, '2026-04-01', d)`	26
`EOMONTH(d, [n])`	`EOMONTH(d)`	2026-04-30
`DATEFROMPARTS(y, m, d)`	`DATEFROMPARTS(2026, 4, 1)`	2026-04-01
`DATETRUNC(part, d)` (2022+)	`DATETRUNC(month, d)`	2026-04-01
`FORMAT(d, 'MMM-yyyy')`		Apr-2026

Date parts you can use: `year`, `quarter`, `month`, `dayofyear`, `day`, `week`, `weekday`, `hour`, `minute`, `second`.

SQL

```sql
SELECT TOP 6
       order_id,
       order_date,
       ship_date,
       YEAR(order_date)                    AS yr,
       DATEPART(quarter, order_date)       AS qtr,
       DATENAME(month, order_date)         AS month_name,
       DATENAME(weekday, order_date)       AS weekday_name,
       DATEDIFF(day, order_date, ship_date) AS days_to_ship,
       EOMONTH(order_date)                 AS month_end
FROM orders
ORDER BY order_date DESC;
```

RESULT

order_id	order_date	ship_date	yr	qtr	month_name	weekday_name	days_to_sh
IN-2025-03197	2025-12-31	2026-01-04	2025	4	December	Wednesday	4
IN-2025-03198	2025-12-31	2026-01-04	2025	4	December	Wednesday	4
IN-2025-03199	2025-12-31	2026-01-02	2025	4	December	Wednesday	2
IN-2025-03200	2025-12-31	2026-01-06	2025	4	December	Wednesday	6
IN-2025-03194	2025-12-30	2026-01-03	2025	4	December	Tuesday	4
IN-2025-03195	2025-12-30	2026-01-06	2025	4	December	Tuesday	7

6 rows × 9 columns

SQL

```sql
SELECT ship_mode,
       COUNT(*)                                      AS orders,
       AVG(DATEDIFF(day, order_date, ship_date) * 1.0)  AS avg_days_to_ship,
       MAX(DATEDIFF(day, order_date, ship_date))        AS max_days
FROM orders
GROUP BY ship_mode
ORDER BY avg_days_to_ship;
```

RESULT

ship_mode	orders	avg_days_to_ship	max_days
Same Day	176	0.00	0
First Class	467	1.53	2
Second Class	621	3.00	4
Standard Class	1936	5.54	7

4 rows × 4 columns

WATCH OUT **DATEDIFF counts boundaries**

`DATEDIFF(year, '2025-12-31', '2026-01-01')` returns **1**, because one year boundary is crossed, even though only one day passed. For exact age use `DATEDIFF(day, …)/365.25` or compare month and day as well.

Useful date patterns

SQL

```sql
-- Orders in the last 30 days
WHERE order_date ⩾ DATEADD(day, -30, CAST(GETDATE() AS DATE))

-- First day of the month
SELECT DATEFROMPARTS(YEAR(order_date), MONTH(order_date), 1)   -- or DATETRUNC(month,
order_date)

-- Indian financial year (April-March)
SELECT CASE WHEN MONTH(order_date) ⩾ 4 THEN YEAR(order_date) + 1 ELSE YEAR(order_date) END
AS fy

-- Month label for reports
SELECT FORMAT(order_date, 'MMM-yyyy') AS month_label
```

Number functions

Function	Example	Result
ROUND(n, d)	ROUND(1234.567, 1)	1234.600
ROUND(n, -2)	ROUND(1234.567, -2)	1200.000
CEILING(n)	CEILING(4.1)	5
FLOOR(n)	FLOOR(4.9)	4
ABS(n)	ABS(-250)	250
POWER(n, p)	POWER(2, 10)	1024
SQRT(n), SQUARE(n)	SQRT(144)	12
SIGN(n)	SIGN(-3)	-1
n % m	17 % 5	2
RAND()		Random float 0–1

SQL

```sql
SELECT sub_category,
       ROUND(SUM(sales), -3)                AS sales_to_thousand,
       CEILING(SUM(quantity) / 12.0)        AS boxes_of_12,
       ROUND(100.0 * SUM(profit) / SUM(sales), 2) AS margin_pct,
       ABS(SUM(profit))                     AS abs_profit,
       SIGN(SUM(profit))                    AS profit_sign
FROM superstore
GROUP BY sub_category
ORDER BY margin_pct;
```

RESULT

sub_category	sales_to_thousand	boxes_of_12	margin_pct	abs_profit	profit_sign
Tables	14,046,000	103	-5.81	816,042	-1
Bookcases	11,275,000	101	-3.75	422,786	-1
Chairs	8,091,000	110	4.23	342,043	1
Furnishings	1,429,000	94	5.50	78,632	1
Copiers	78,750,000	99	10.71	8,436,141	1
Phones	31,919,000	110	11.48	3,663,117	1
Accessories	3,179,000	98	11.63	369,554	1
Machines	42,590,000	106	12.18	5,186,711	1
Art	246,000	103	14.85	36,526	1
Appliances	6,334,000	106	14.92	944,927	1
Storage	2,659,000	107	14.94	397,228	1
Paper	493,000	124	15.21	75,040	1

13 rows × 6 columns (first 12 shown)

Converting types: CAST, CONVERT, TRY_CAST

Function	Example	Notes
`CAST(x AS type)`	`CAST('2026-04-27' AS DATE)`	ANSI standard
`CONVERT(type, x, style)`	`CONVERT(VARCHAR(10), order_date, 103)` → 27/04/2026	SQL Server; style codes format dates (103 = dd/mm/yyyy, 23 = yyyy-mm-dd)
`TRY_CAST(x AS type)`	`TRY_CAST('abc' AS INT)` → NULL	Returns NULL instead of an error: great for dirty data
`TRY_CONVERT`, `TRY_PARSE`		Same idea

SQL

```sql
-- Find rows where a text column does not contain a valid number
SELECT * FROM staging_sales WHERE amount_text IS NOT NULL AND TRY_CAST(amount_text AS DECIMAL(12,2)) IS NULL;
```

Conditional functions (preview)

Function	Example
`COALESCE(a, b, …)`	`COALESCE(phone, email, 'no contact')`
`ISNULL(a, b)`	`ISNULL(discount, 0)`
`NULLIF(a, b)`	`profit / NULLIF(sales, 0)`
`IIF(cond, a, b)`	`IIF(profit < 0, 'Loss', 'Profit')`
`CASE WHEN … END`	Multi-branch logic (Lesson 7)

SQL

```sql
SELECT TOP 6 order_id, sales, profit,
       IIF(profit < 0, 'Loss', 'Profit')          AS result,
       ROUND(profit / NULLIF(sales, 0) * 100, 1)  AS margin_pct
FROM superstore
ORDER BY profit;
```

RESULT

order_id	sales	profit	result	margin_pct
IN-2025-03027	318,492	-75,874	Loss	-23.8
IN-2023-00320	337,316	-67,393	Loss	-20.0
IN-2023-00372	378,192	-59,796	Loss	-15.8
IN-2024-01335	194,160	-56,392	Loss	-29.0
IN-2025-02299	202,475	-46,375	Loss	-22.9
IN-2023-00077	240,940	-45,712	Loss	-19.0

6 rows × 5 columns

DO IT WITH AI **Do it with AI**

Prompt: *"In SQL Server, write a query on superstore(order_id, order_date, ship_date, ship_mode) that returns the average and 90th percentile of days to ship by ship mode and quarter."* Expect it to use `DATEDIFF`, `DATEPART(quarter, …)` and `PERCENTILE_CONT(0.9) WITHIN GROUP (ORDER BY …) OVER (PARTITION BY …)`, which you will meet with window functions in Lesson 12.

Summary

KEY TAKEAWAYS

- **Scalar** functions work per row; **aggregate** functions summarise many rows.
- `COUNT(*)` counts rows; `COUNT(col)` skips NULLs; `COUNT(DISTINCT col)` counts unique values; AVG ignores NULLs.
- Text: `LEN`, `LEFT/RIGHT/SUBSTRING`, `CHARINDEX`, `REPLACE`, `TRIM`, `UPPER/LOWER`, `CONCAT`, `STRING_AGG`, `STRING_SPLIT`.
- Dates: `YEAR/MONTH/DAY`, `DATEPART`, `DATENAME`, `DATEADD`, `DATEDIFF` (counts boundaries), `EOMONTH`, `DATEFROMPARTS`, `DATETRUNC`.
- Numbers: `ROUND` (negative digits round to tens/hundreds), `CEILING`, `FLOOR`, `ABS`, `POWER`, `SQRT`, `%`.
- Convert with `CAST/CONVERT`; use `TRY_CAST` on dirty data.

Quick quiz

6 questions · tap an answer to check it

1. A column has values 10, NULL, 20. What does `AVG(col)` return?

A 10

B 15

C 30

D NULL

Answer: B. AVG ignores NULLs: (10 + 20) / 2 = 15.

2. What does `SUBSTRING('IN-2025-00042', 4, 4)` return?

A IN-2

B 2025

C 0004

D 5-00

Answer: B. Starting at position 4, take 4 characters.

3. Which returns the number of days between order and ship dates?

A `DATEPART(day, ship_date - order_date)`

B `DATEDIFF(day, order_date, ship_date)`

C `DATEADD(day, order_date, ship_date)`

D `DAY(ship_date) - DAY(order_date)`

Answer: B. DATEDIFF(datepart, start, end) counts the boundaries between the two dates.

4. What does `ROUND(45678.9, -3)` return?

A 45678.000

B 45679

C 46000.0

D 45000

Answer: C. −3 rounds to the nearest thousand.

5. Which function returns NULL instead of an error when text cannot be converted?

A CAST

B CONVERT

C TRY_CAST

D ISNULL

Answer: C. TRY_CAST and TRY_CONVERT return NULL on conversion failure.

6. Which counts unique customers?

A `COUNT(customer_id)`

B `COUNT(DISTINCT customer_id)`

C `SUM(DISTINCT customer_id)`

D `DISTINCT COUNT(customer_id)`

Answer: B. COUNT(DISTINCT ...) counts distinct non-NULL values.

EXERCISE **Function practice on Superstore**

Write one query that returns, for each order year: the number of orders, the number of distinct customers, total sales rounded to the nearest lakh (100,000), the average days to ship, and a comma-separated list of the ship modes used.

Show solution (worked answer)

SQL

```sql
WITH modes AS (                                  -- one row per year and ship mode
    SELECT DISTINCT YEAR(order_date) AS order_year, ship_mode FROM orders
),
mode_list AS (
    SELECT order_year, STRING_AGG(ship_mode, ', ') AS ship_modes FROM modes GROUP BY
order_year
)
SELECT YEAR(o.order_date)                                    AS order_year,
       COUNT(DISTINCT o.order_id)                            AS orders,
       COUNT(DISTINCT o.customer_id)                         AS customers,
       ROUND(SUM(oi.sales), -5)                              AS sales_nearest_lakh,
       AVG(DATEDIFF(day, o.order_date, o.ship_date) * 1.0)   AS avg_days_to_ship,
       MAX(m.ship_modes)                                     AS ship_modes
FROM orders AS o
JOIN order_items AS oi ON oi.order_id = o.order_id
JOIN mode_list   AS m  ON m.order_year = YEAR(o.order_date)
GROUP BY YEAR(o.order_date)
ORDER BY order_year;
```

`STRING_AGG(DISTINCT …)` is not allowed in SQL Server, so the distinct ship modes are collected in a CTE first (CTEs are covered in Lesson 13). Note that averaging days to ship over order *lines* weights orders with more items more heavily; average over `orders` alone if each order should count once.

INTERVIEW CORNER

Questions you may be asked

Q: COUNT() vs COUNT(column) vs COUNT(DISTINCT column)? COUNT() counts rows; COUNT(column) counts non-NULL values; COUNT(DISTINCT column) counts unique non-NULL values.

Q: How do you extract the domain from an email in SQL Server? `SUBSTRING(email, CHARINDEX('@', email) + 1, LEN(email))` or `RIGHT(email, LEN(email) - CHARINDEX('@', email))`.

Q: How do you get the last day of the previous month? `EOMONTH(GETDATE(), -1)`.

Q: Difference between CAST and CONVERT? CAST is ANSI standard; CONVERT is SQL Server-specific and accepts a style argument for date/number formatting.

Practise more online

Free, trusted places to test yourself on this topic. Links open the original site in a new tab.